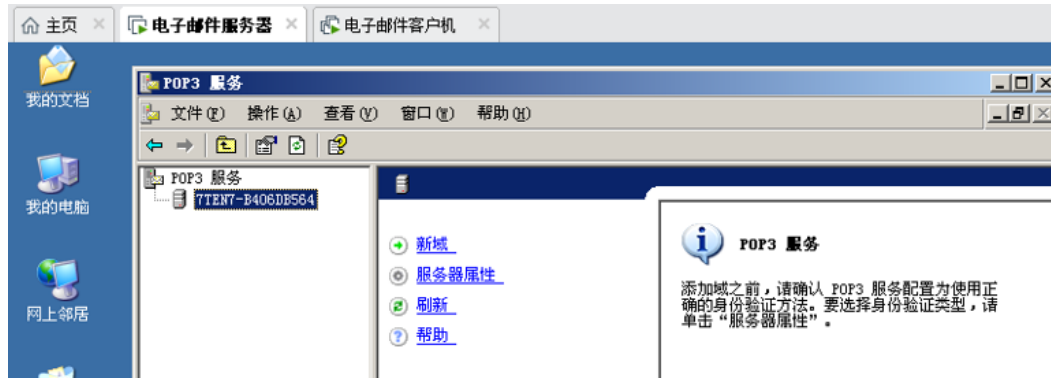


注意：由于实验环境特殊平台暂不支持，所以仅提供邮件流量包进行分析，以下内容为流量包的抓取过程与分析过程

一、实验步骤

1. 准备一台电子邮件服务器和一台电子邮件客户机



2. 查看服务器与客户机的 IP 地址

服务器：

```
C:\Documents and Settings\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter 本地连接:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.247.100
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.247.2
```

客户机：

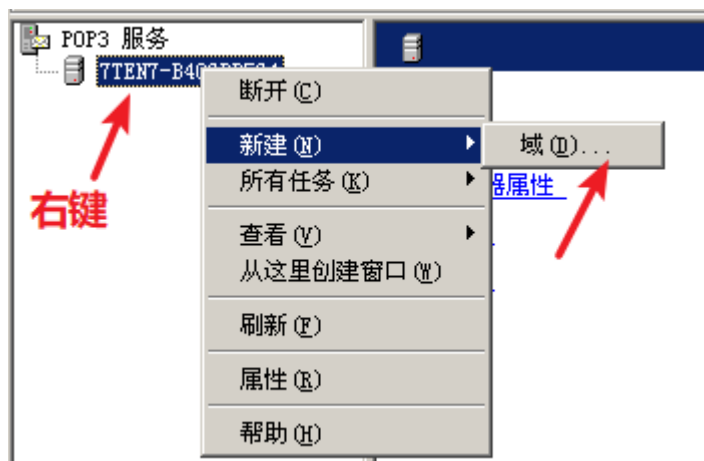
```
C:\Documents and Settings\Administrator>ipconfig

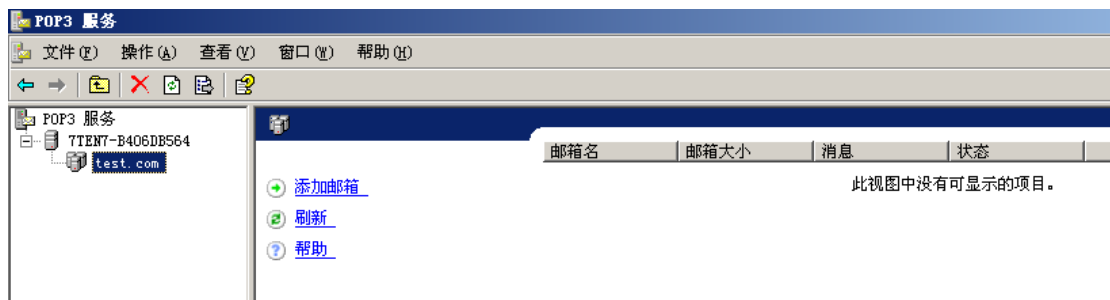
Windows IP Configuration

Ethernet adapter 本地连接:

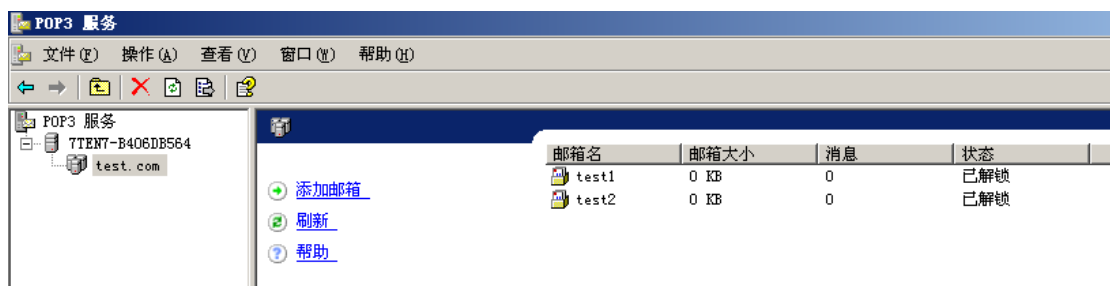
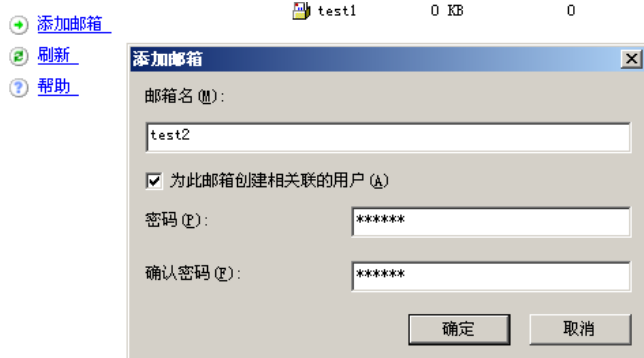
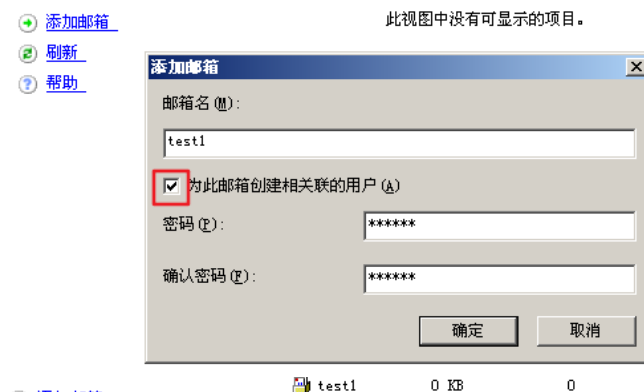
    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.247.200
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.247.2
```

3. 在服务器的 pop3 服务中新建一个 test.com 域

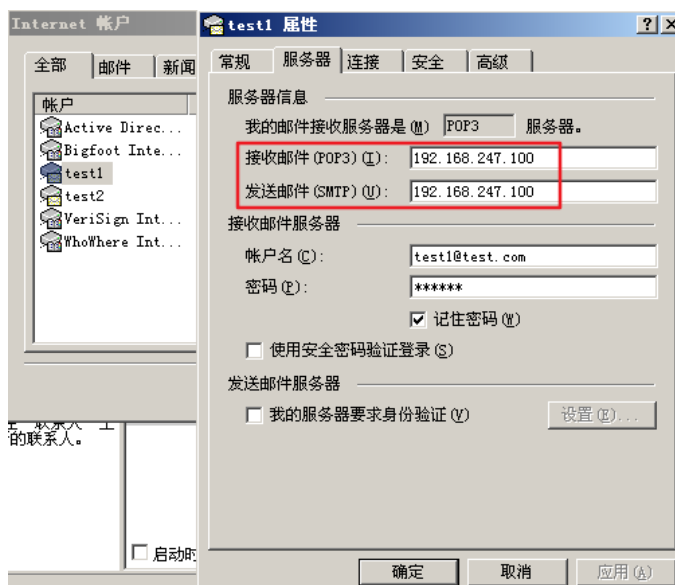
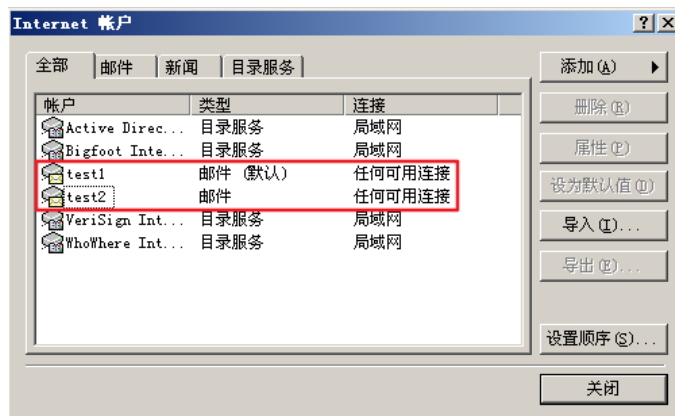
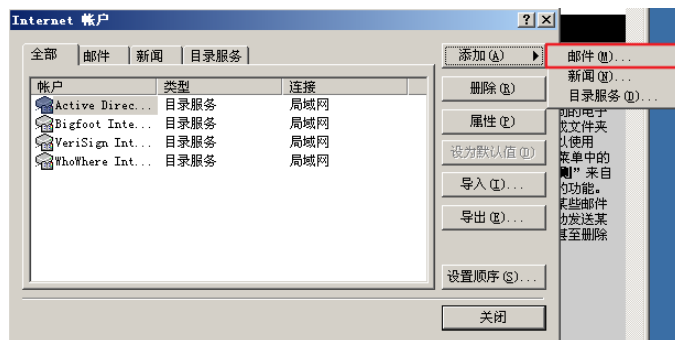
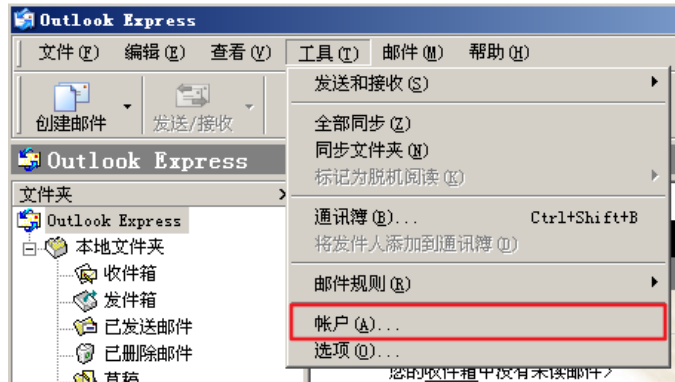




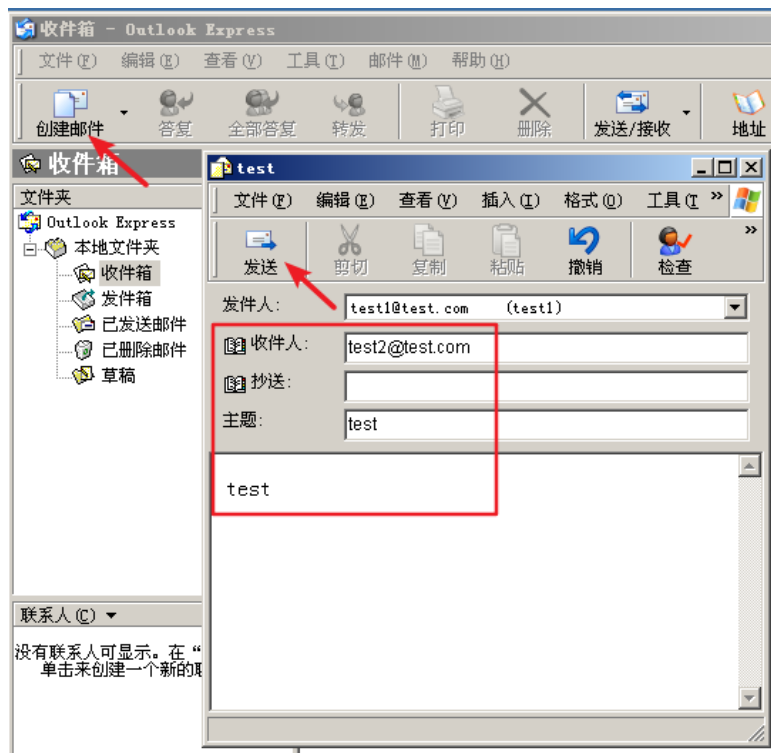
4. 在域中添加两个邮箱，用户名分别为 test1 和 test2，密码均为 123123



5. 在客户机中打开 outlook，添加账户 test1 与 test2
接收邮件服务器地址 与 发送邮件服务器地址 都为服务器 1 的 IP 地址



6. 在客户机中使用账户 test1 向账户 test2 发送邮件，并抓包



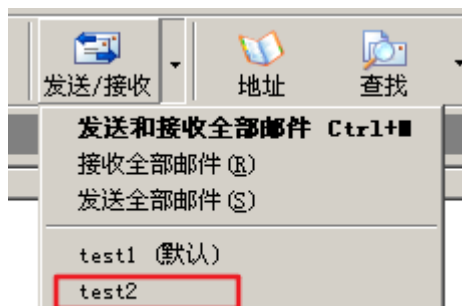
smtp.pcapng [Wireshark 1.10.0 (SVN Rev 49790 from /trunk-1.10)]

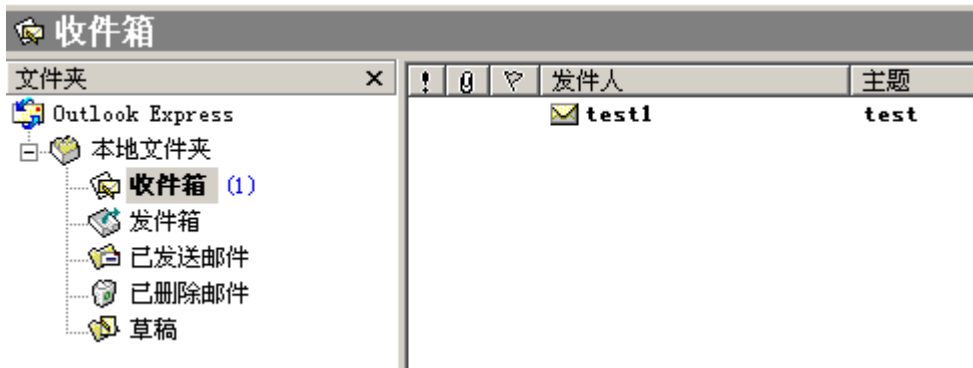
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help

Filter: Expression... Clear Apply Save

No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000000	192.168.247.200	192.168.247.100	TCP	62	ssslg-mgr > smtp [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PER
2	0.00025400	192.168.247.100	192.168.247.200	TCP	62	smtp > ssslg-mgr [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=14
3	0.00027300	192.168.247.200	192.168.247.100	TCP	54	ssslg-mgr > smtp [ACK] Seq=1 Ack=1 Win=64240 Len=0
4	0.00046100	192.168.247.100	192.168.247.200	SMTP	170	S: 220 7ten7-b406db564 Microsoft ESMTMP MAIL Service, version: 6
5	0.00066700	192.168.247.100	192.168.247.200	SMTP	75	C: HELO 7ten7-b406db564
6	0.00084700	192.168.247.100	192.168.247.200	SMTP	99	S: 250 7ten7-b406db564 Hello [192.168.247.200]
7	0.00101600	192.168.247.200	192.168.247.100	SMTP	83	C: MAIL FROM: <test1@test.com>
8	0.00137800	192.168.247.100	192.168.247.200	SMTP	93	S: 250 2.1.0 test1@test.com...Sender OK
9	0.00142800	192.168.247.200	192.168.247.100	SMTP	81	C: RCPT TO: <test2@test.com>
10	0.00203800	192.168.247.100	192.168.247.200	SMTP	81	S: 250 2.1.5 test2@test.com
11	0.00249000	192.168.247.200	192.168.247.100	SMTP	60	C: DATA
12	0.00519900	192.168.247.100	192.168.247.200	SMTP	100	S: 354 Start mail input; end with <CRLF>.<CRLF>
13	0.00535300	192.168.247.200	192.168.247.100	SMTP	488	C: DATA fragment, 434 bytes
14	0.15467300	192.168.247.100	192.168.247.200	TCP	60	smtp > ssslg-mgr [ACK] Seq=274 Ack=518 Win=63723 Len=0
15	0.15470800	192.168.247.200	192.168.247.100	IMF	59	From: 'test1' <test1@test.com>, Subject: test, (text/plain)
16	0.15574500	192.168.247.200	192.168.247.100	SMTP	141	S: 250 2.6.0 <B6691E8A6D0540C09F012B762498E94967ten7b406db564>
17	0.15582600	192.168.247.200	192.168.247.100	SMTP	60	C: QUIT
18	0.15604000	192.168.247.100	192.168.247.200	SMTP	118	S: 221 2.0.0 7ten7-b406db564 Service closing transmission chan
19	0.15608500	192.168.247.200	192.168.247.100	TCP	54	ssslg-mgr > smtp [FIN, ACK] Seq=529 Ack=425 Win=63816 Len=0
20	0.15725900	192.168.247.100	192.168.247.200	TCP	60	smtp > ssslg-mgr [FIN, ACK] Seq=425 Ack=529 Win=63712 Len=0
21	0.15726700	192.168.247.200	192.168.247.100	TCP	54	ssslg-mgr > smtp [ACK] Seq=530 Ack=426 Win=63816 Len=0
22	0.15729800	192.168.247.100	192.168.247.200	TCP	60	smtp > ssslg-mgr [ACK] Seq=426 Ack=530 Win=63712 Len=0

7. 在客户机中切换账户为 test2 进行邮件接收，并抓包





pop.pcapng [Wireshark 1.10.0 (SVN Rev 49790 from /trunk-1.10)]						
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help						
Filter: Expression... Clear Apply Save						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.00000000	192.168.247.200	192.168.247.100	TCP	62	accord-mgc > pop3 [SYN] Seq=0 win=64240 Len=0 MSS=1460 SACK_PERM=1
2	0.00026300	192.168.247.100	192.168.247.200	TCP	62	pop3 > accord-mgc [SYN, ACK] Seq=0 Ack=1 win=64240 Len=0 MSS=1460
3	0.00028100	192.168.247.200	192.168.247.100	TCP	54	accord-mgc > pop3 [ACK] Seq=1 Ack=1 win=64240 Len=0
4	0.00060000	192.168.247.100	192.168.247.200	POP	136	S: +OK Microsoft windows POP3 Service Version 1.0 <1156828107t@163.com>
5	0.00085000	192.168.247.200	192.168.247.100	POP	75	C: USER test2@test.com
6	0.00111300	192.168.247.100	192.168.247.200	POP	60	S: +OK
7	0.00114800	192.168.247.200	192.168.247.100	POP	67	C: PASS 123123
8	0.00179600	192.168.247.100	192.168.247.200	POP	87	S: +OK User successfully logged on
9	0.00204700	192.168.247.200	192.168.247.100	POP	60	C: STAT
10	0.00229900	192.168.247.100	192.168.247.200	POP	65	S: +OK 1 693
11	0.00256100	192.168.247.200	192.168.247.100	POP	60	C: LIST
12	0.00277300	192.168.247.100	192.168.247.200	POP	93	S: +OK 1 messages (693 octets)
13	0.00332900	192.168.247.200	192.168.247.100	POP	62	C: RETR 1
14	0.00361200	192.168.247.100	192.168.247.200	POP	769	S: +OK 693 octets
15	0.00456000	192.168.247.200	192.168.247.100	POP	62	C: DELE 1
16	0.00500700	192.168.247.100	192.168.247.200	POP	85	S: +OK Message marked as deleted
17	0.00539900	192.168.247.200	192.168.247.100	POP	60	C: QUIT
18	0.00585700	192.168.247.100	192.168.247.200	POP	142	S: +OK Microsoft windows POP3 Service Version 1.0 <1156828107t@163.com>
19	0.00586200	192.168.247.100	192.168.247.200	TCP	60	pop3 > accord-mgc [FIN, ACK] Seq=1005 Ack=69 win=64172 Len=0
20	0.00586700	192.168.247.200	192.168.247.100	TCP	54	accord-mgc > pop3 [ACK] Seq=69 Ack=1006 win=63236 Len=0
21	0.00595600	192.168.247.200	192.168.247.100	TCP	54	accord-mgc > pop3 [FIN, ACK] Seq=69 Ack=1006 win=63236 Len=0
22	0.00611300	192.168.247.100	192.168.247.200	TCP	60	pop3 > accord-mgc [ACK] Seq=1006 Ack=70 win=64172 Len=0

二、 SMTP 流量分析

1. 客户机通过 TCP 三次握手和服务器建立了连接

1	0.000000	192.168.247.200	192.168.247.100	TCP	62	1204 → 25 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1
2	0.000254	192.168.247.100	192.168.247.200	TCP	62	25 → 1204 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1
3	0.000273	192.168.247.200	192.168.247.100	TCP	54	1204 → 25 [ACK] Seq=1 Ack=1 Win=64240 Len=0

2. 服务器返回 220 表示服务就绪

4	0.000461	192.168.247.100	192.168.247.200	SMTP	170	S: 220 7ten7-b406db564
---	----------	-----------------	-----------------	------	-----	------------------------

3. 客户机向服务器表明身份,

5	0.000667	192.168.247.200	192.168.247.100	SMTP	75	C: HELO 7ten7b406db564
6	0.000847	192.168.247.100	192.168.247.200	SMTP	99	S: 250 7ten7-b406db564 Hello [192.168.247.200]

4. 客户机告诉服务器谁是邮件的发送者

7	0.001016	192.168.247.200	192.168.247.100	SMTP	83	C: MAIL FROM: <test1@test.com>
8	0.001378	192.168.247.100	192.168.247.200	SMTP	93	S: 250 2.1.0 test1@test.com....Sender OK

5. 客户机告诉服务器谁是邮件的接收者

9	0.001428	192.168.247.200	192.168.247.100	SMTP	81	C: RCPT TO: <test2@test.com>
10	0.002038	192.168.247.100	192.168.247.200	SMTP	81	S: 250 2.1.5 test2@test.com

6. 11-客户机请求发送数据

12-服务器返回 354 表示等待客户机输入，并且表示将以换行符（<CRLF>.<CRLF>）作为数据的结尾

13-客户机向服务器发送了 434 字节的数据

14-服务器向客户机返回了数据接收完成所用的时间

11 0.002490	192.168.247.200	192.168.247.100	SMTP	60 C: DATA
12 0.005199	192.168.247.100	192.168.247.200	SMTP	100 S: [354] Start mail input; end with [CRLF]>[CRLF]
13 0.005353	192.168.247.200	192.168.247.100	SMTP	488 C: DATA fragment, [434 bytes]
14 0.154673	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [ACK] Seq=274 Ack=518 Win=63723 Len=0
15 0.154709	192.168.247.200	192.168.247.100	SMTP/I...	59 from: "test1" <test1@test.com>, subject: test, (text
16 0.155745	192.168.247.100	192.168.247.200	SMTP	141 S: 250 2.6.0 <B6691E8A6D0540C09F012B762498E949@7ten7
17 0.155826	192.168.247.200	192.168.247.100	SMTP	60 C: QUIT
18 0.156040	192.168.247.100	192.168.247.200	SMTP	118 S: 221 2.0.0 7ten7-b406db564 Service closing transmis
19 0.156085	192.168.247.200	192.168.247.100	TCP	54 1204 → 25 [FIN, ACK] Seq=529 Ack=425 Win=63816 Len=0
20 0.157259	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [FIN, ACK] Seq=425 Ack=529 Win=63712 Len=0
21 0.157267	192.168.247.200	192.168.247.100	TCP	54 1204 → 25 [ACK] Seq=530 Ack=426 Win=63816 Len=0
22 0.157298	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [ACK] Seq=426 Ack=530 Win=63712 Len=0

[Checksum Status: Unverified]				
Urgent pointer: 0				
> [SEQ/ACK analysis]				
v [Timestamps]				
[Time since first frame in this TCP stream: 0.154673000 seconds]				
[Time since previous frame in this TCP stream: 0.149320000 seconds]				
0000 00 0c 29 fc ca 64 00 0c 29 b7 31 c4 08 00 45 00 --)·d· ·)·1··E·				
0010 00 28 0b db 40 00 00 06 7e 76 c0 a8 f7 64 c0 a8 ·(·@····~v··d· ·				
0020 f7 c8 00 19 04 b4 b9 94 30 82 74 94 5c 44 50 10 ······0·t·\DP·				
0030 f8 eb 86 ad 00 00 00 00 00 00 00 00 00 00 00 00 ······ ····				

7. 客户机向服务器询问邮件状态，服务器表示邮件已进入投递队列（Queued mail for delivery）

15 0.154709	192.168.247.200	192.168.247.100	SMTP/I...	59 from: "test1" <test1@test.com>, subject: test, (text/plain)
16 0.155745	192.168.247.100	192.168.247.200	SMTP	141 S: 250 2.6.0 <B6691E8A6D0540C09F012B762498E949@7ten7b406db564> Queued mail for delivery
17 0.155826	192.168.247.200	192.168.247.100	SMTP	60 C: QUIT
18 0.156040	192.168.247.100	192.168.247.200	SMTP	118 S: 221 2.0.0 7ten7-b406db564 Service closing transmission channel
19 0.156085	192.168.247.200	192.168.247.100	TCP	54 1204 → 25 [FIN, ACK] Seq=529 Ack=425 Win=63816 Len=0
20 0.157259	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [FIN, ACK] Seq=425 Ack=529 Win=63712 Len=0
21 0.157267	192.168.247.200	192.168.247.100	TCP	54 1204 → 25 [ACK] Seq=530 Ack=426 Win=63816 Len=0
22 0.157298	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [ACK] Seq=426 Ack=530 Win=63712 Len=0

[Time since previous frame in this TCP stream: 0.001036000 seconds]				
TCP payload (87 bytes)				
v Simple Mail Transfer Protocol				
v Response: 250 2.6.0 <B6691E8A6D0540C09F012B762498E949@7ten7b406db564> Queued mail for delivery\r\n				
Response code: Requested mail action okay, completed (250)				
Response parameter: 2.6.0 <B6691E8A6D0540C09F012B762498E949@7ten7b406db564> Queued mail for delivery				

8. 客户机发起退出邮件服务请求，服务器返回 221 表示服务结束

17 0.155826	192.168.247.200	192.168.247.100	SMTP	60 C: QUIT
18 0.156040	192.168.247.100	192.168.247.200	SMTP	118 S: 221 2.0.0 7ten7-b406db564 Service closing transmission channel

9. 服务器与客户机经过 TCP 四次挥手后断开连接

19 0.156085	192.168.247.200	192.168.247.100	TCP	54 1204 → 25 [FIN, ACK] Seq=529 Ack=425 Win=63816 Len=0
20 0.157259	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [FIN, ACK] Seq=425 Ack=529 Win=63712 Len=0
21 0.157267	192.168.247.200	192.168.247.100	TCP	54 1204 → 25 [ACK] Seq=530 Ack=426 Win=63816 Len=0
22 0.157298	192.168.247.100	192.168.247.200	TCP	60 25 → 1204 [ACK] Seq=426 Ack=530 Win=63712 Len=0

三、 POP3 流量分析

1. 客户机通过 TCP 三次握手和服务器建立了连接

1 0.000000	192.168.247.200	192.168.247.100	TCP	62 1205 → 110 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1
2 0.000263	192.168.247.100	192.168.247.200	TCP	62 110 → 1205 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM=1
3 0.000281	192.168.247.200	192.168.247.100	TCP	54 1205 → 110 [ACK] Seq=1 Ack=1 Win=64240 Len=0

2. 服务器表示 POP3 协议已准备就绪，版本为 1.0

4 0.000600	192.168.247.100	192.168.247.200	POP	136 S: +OK Microsoft Windows POP3 Service Version 1.0 <11568281@7ten7-b406db564> ready.
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3. 客户机登录 test2 邮箱，准备接收邮件

5 0.000850	192.168.247.200	192.168.247.100	POP	75 C: USER test2@test.com
6 0.001113	192.168.247.100	192.168.247.200	POP	60 S: +OK
7 0.001148	192.168.247.200	192.168.247.100	POP	67 C: PASS 123123
8 0.001796	192.168.247.100	192.168.247.200	POP	87 S: +OK User successfully logged on

4. 客户机请求服务器发回关于邮箱的统计资料，服务器返回邮件共 1 封，共 693 字节

9 0.002047	192.168.247.200	192.168.247.100	POP	60 C: STAT
10 0.002299	192.168.247.100	192.168.247.200	POP	65 S: +OK 1 693

5. 客户机让服务器返回邮件数量和每个邮件的大小，服务器表示只有一封邮件，大小为 693 字节

11 0.002561	192.168.247.200	192.168.247.100	POP	60 C: LIST
12 0.002773	192.168.247.100	192.168.247.200	POP	93 S: +OK 1 messages (693 octets)

6. 客户机请求服务器返回该邮件的全部文本，RETR 1 表示只接收一封邮件，服务器返回

OK 表示完成，邮件大小共 693 字节

13	0.003329	192.168.247.200	192.168.247.100	POP	62 C: RETR 1
14	0.003612	192.168.247.100	192.168.247.200	POP	769 S: +OK 693 octects

7. 客户机让服务器从投递列表中删除一封邮件

15	0.004560	192.168.247.200	192.168.247.100	POP	62 C: DELE 1
16	0.005007	192.168.247.100	192.168.247.200	POP	85 S: +OK Message marked as deleted

8. 客户机发起退出服务请求

17	0.005399	192.168.247.200	192.168.247.100	POP	60 C: QUIT
18	0.005857	192.168.247.100	192.168.247.200	POP	142 S: +OK Microsoft Windows POP3 Service Version 1.0 <11568281@ten7-b406db564> signing off.

9. 服务器和客户机通过 TCP 四次挥手断开连接

19	0.005862	192.168.247.100	192.168.247.200	TCP	60 110 → 1205 [FIN, ACK] Seq=1005 Ack=69 Win=64172 Len=0
20	0.005867	192.168.247.200	192.168.247.100	TCP	54 1205 → 110 [ACK] Seq=69 Ack=1006 Win=63236 Len=0
21	0.005956	192.168.247.200	192.168.247.100	TCP	54 1205 → 110 [FIN, ACK] Seq=69 Ack=1006 Win=63236 Len=0
22	0.006113	192.168.247.100	192.168.247.200	TCP	60 110 → 1205 [ACK] Seq=1006 Ack=70 Win=64172 Len=0

四、安全隐患

通过观察邮件发送/接收的数据包，可以发现邮件的加密方式为简单的 base64 编码，未采用签名算法。若邮件被恶意抓包，容易导致信息泄露。

14	0.003612	192.168.247.100	192.168.247.200	POP	769 S: +OK 693 octects
15	0.004560	192.168.247.200	192.168.247.100	POP	62 C: DELE 1
16	0.005007	192.168.247.100	192.168.247.200	POP	85 S: +OK Message marked
17	0.005399	192.168.247.200	192.168.247.100	POP	60 C: QUIT
18	0.005857	192.168.247.100	192.168.247.200	POP	142 S: +OK Microsoft Wind
19	0.005862	192.168.247.100	192.168.247.200	TCP	60 110 → 1205 [FIN, ACK]
20	0.005867	192.168.247.200	192.168.247.100	TCP	54 1205 → 110 [ACK] Seq=
21	0.005956	192.168.247.200	192.168.247.100	TCP	54 1205 → 110 [FIN, ACK]
22	0.006113	192.168.247.100	192.168.247.200	TCP	60 110 → 1205 [ACK] Seq=

```
X-OriginalArrivalTime: 01 Nov 2019 13:04:06.0328 (UTC) FILETIME=[D79E3380:01D590B4]\r\n\r\n
dGVzdA==\r\n\r\n
\r\n
\r\n
.\r\n
```

0220	74 20 4f 75 74 6c 6f 6f 6b 20 45 78 70 72 65 73	t Outloo k Expres
0230	73 20 36 2e 30 30 2e 33 37 39 30 2e 33 39 35 39	s 6.00.3 790.3959
0240	0d 0a 58 2d 4d 69 6d 65 4f 4c 45 3a 20 50 72 6f	·X-Mime OLE: Pro
0250	64 75 63 65 64 20 42 79 20 4d 69 63 72 6f 73 6f	duced By Microso
0260	66 74 20 4d 69 6d 65 4f 4c 45 20 56 36 2e 30 30	ft Mime0 LE V6.00
0270	2e 33 37 39 30 2e 33 39 35 39 0d 0a 52 65 74 75	.3790.39 59·Retu
0280	72 6e 2d 50 61 74 68 3a 20 74 65 73 74 31 40 74	rn-Path: test1@t
0290	65 73 74 2e 63 6f 6d 0d 0a 58 2d 4f 72 69 67 69	est.com·X-Orig
02a0	6e 61 6c 41 72 72 69 76 61 6c 54 69 6d 65 3a 20	nalArriv alTime:
02b0	30 31 20 4e 6f 76 20 32 30 31 39 20 31 33 3a 30	01 Nov 2 019 13:0
02c0	34 3a 30 36 2e 30 33 32 38 20 28 55 54 43 29 20	4:06.032 8 (UTC)
02d0	46 49 4c 45 54 49 4d 45 3d 5b 44 37 39 45 33 33	FILETIME =[D79E33
02e0	38 30 3a 30 31 44 35 39 30 42 34 5d 0d 0a 0d 0a	80:01D59 0B4]....
02f0	64 47 56 7a 64 41 3d 3d 0d 0a 0d 0a 0d 0a 2e 0d	dGVzdA== test
0300	0a	.